

Digital Circuits

Exercise 1- Digital Building Block Measurements

Group/Team No.:

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1st lab exercise	max score	score reached
Security training	mandatory	0
Preparation	0	0
Task 1 (static behavior)	2	
Task 2.1 (generator adjustment)	2	
Task 2.2 (rise/fall time, delay)	3	
Task 2.3 (delay)	1	
Task 3.1 (glitch)	3	
Task 4.1 (oscillator)	2	
Sum	13	
Protocol	5	
Penalty		
Total points	18	

Task 1: Behavior of a gate with static power supply

Introduction

The purpose of the exercise is to analyze the behavior of one NAND gate inside the 7400 IC. Two switches are connected to an LED for this purpose.

On the other hand, safety instructions are given, regarding powering limits of the Evaluation Board.



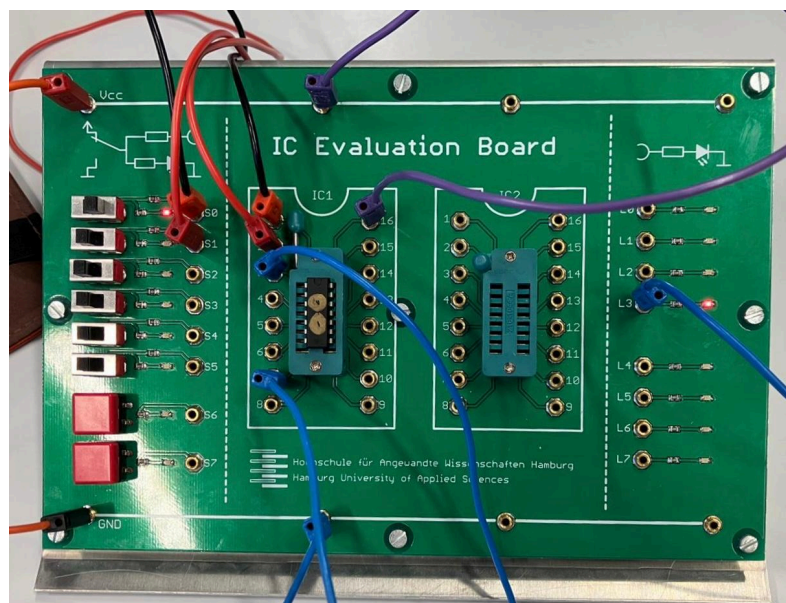
Schematic setup of Task 1

Procedure

1. A DC source of 3.3 V is connected to power the Evaluation Board and the 7400 IC.
2. Two switches are connected individually to the inputs of one NAND gate.
3. An LED is connected to the output of the logic gate.
4. Different behaviors are visible for different switch positions; these results are presented in the report.

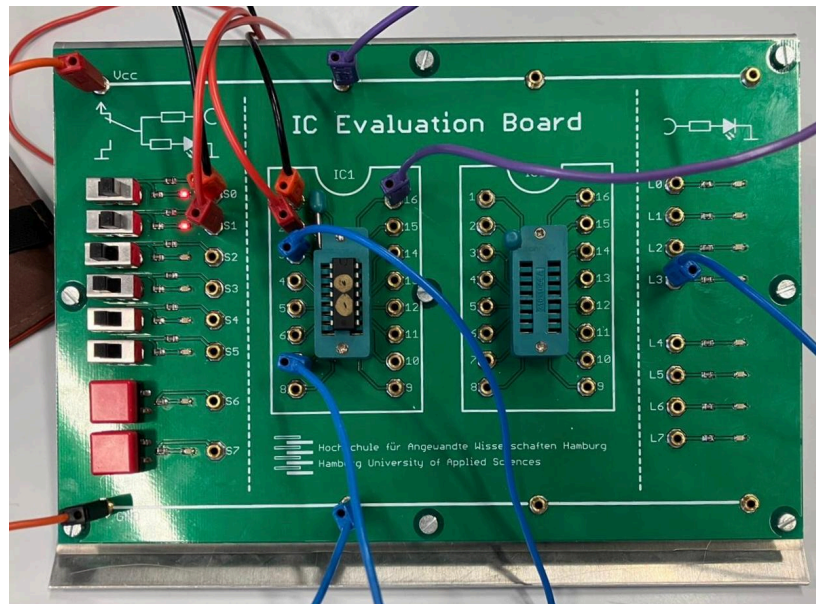
The following configuration is chosen for the first test:

- Input pin 1: ON
- Input pin 2: OFF
- Output LED: ON



The following configuration is chosen for the second test:

- Input pin 1: ON
- Input pin 2: ON
- Output LED: OFF



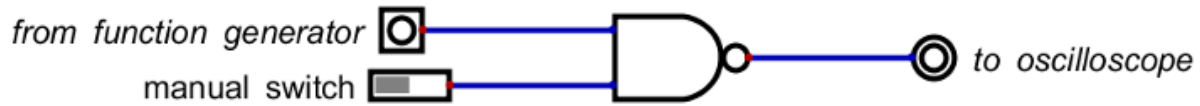
Conclusion

The results are aligned with the expectations of a NAND gate. The exercise provides insights of how to operate laboratory equipment in the digital circuits laboratories, such as specific oscilloscopes, probes, and Evaluation Boards.

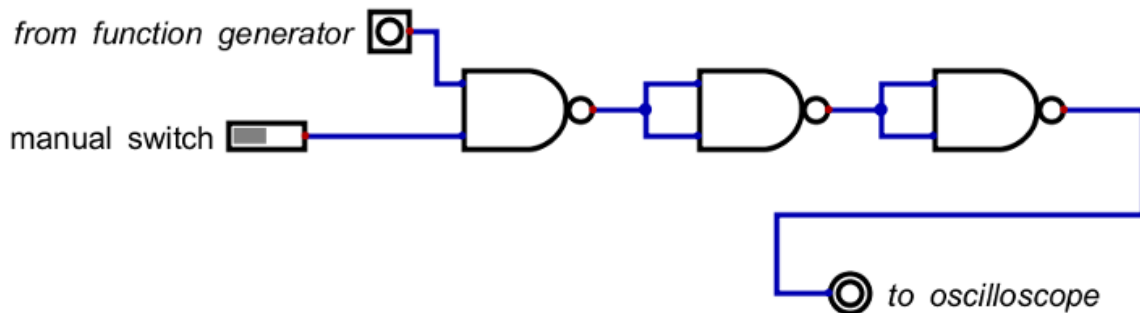
Task 2: Timing characteristic of a gate

Introduction

The objective of the tasks is successfully setting up a function generator to produce a square signal. Furthermore, the rise and fall time is inspected for a single logic gate, connecting the function generator to one of the inputs.



Schematic setup of Task 2.1



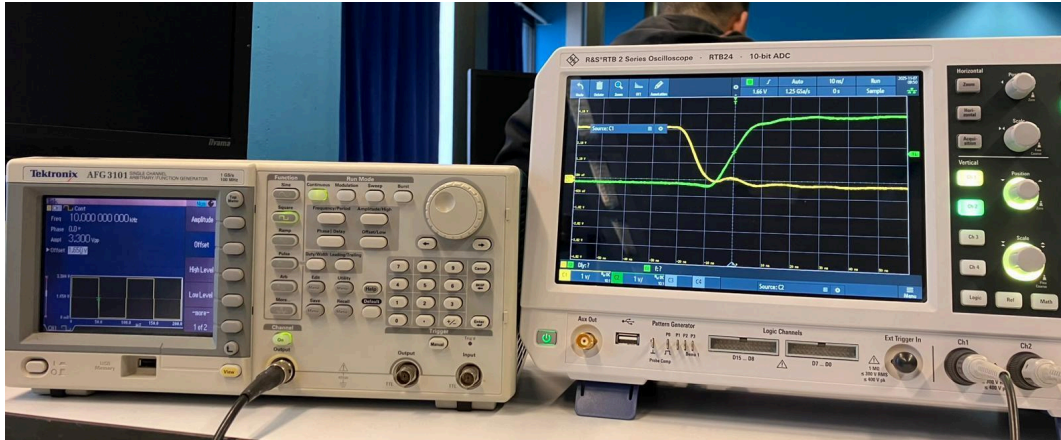
Schematic setup of Task 2.3

Procedure

A function generator is configured to produce a square wave with the following characteristics:

- Frequency: 10k Hz
- Amplitude: 3.3 Vpp
- Offset: 1.65 V

The NAND is wired following the provided schematic setup of Task 2.1. The oscilloscope records the raw function generator signal in Channel 1 --displayed in yellow--, and the NAND gate output in Channel 2 --displayed in green.



Function generator and oscilloscope setup



Full period observation



Rise time observation



Fall time observation

Based on the results obtained with the oscilloscope, the switching characteristics of both signals are analyzed. Under a stimulus with a rising time of 10 ns, the logic gate reacts with a fall time of 30 ns. On the other hand, while the stimulus signal falls in 10 ns, the logic gate rises in 20 ns.

Task 2.3: Propagation of delay - Procedure

The rise and fall times of the signal generator are observed to be 10 ns. However, due to the additional logic gates involved in this task, a rise time of 50 ns, and a fall time of 30 ns are registered.



Rising time observation



Fall time observation

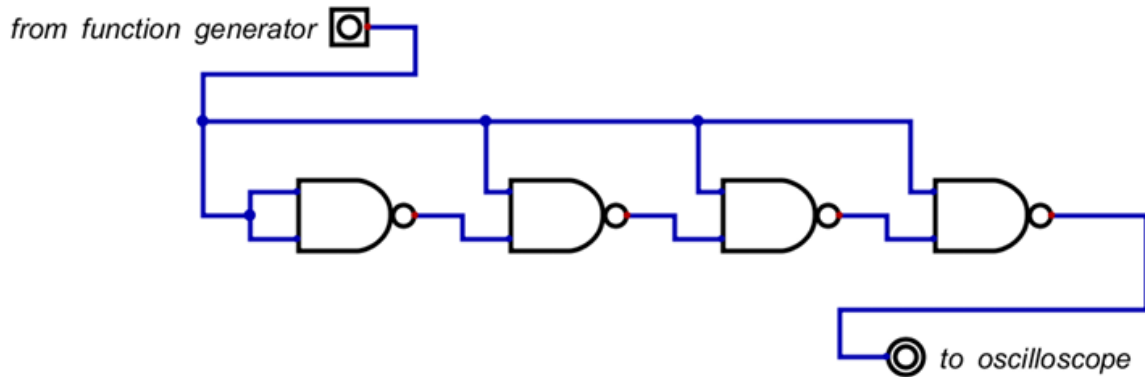
Conclusion

There's a clear difference between the delays occurring during fall and rise, and that can be explained by the internal PNP and NPN transistors conforming the CMOS. Assuming that the manufacturer of the IC 7400 chose arbitrary transistors, this would lead to an asymmetry due to the electron and hole speed differences. Additionally, the amount of wires involved in the setup have a possible influence in the delays generated by the setups --compared to integrated ICs that minimize this effects.

Task 3: Glitch Observation

Introduction

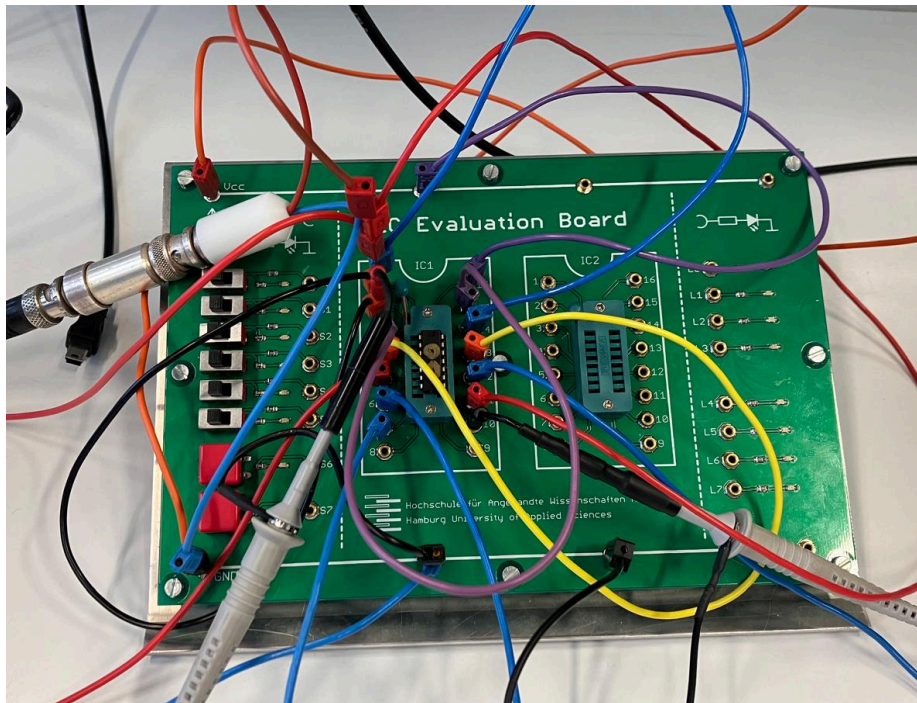
Multiple logic gates are now connected following the schematic of the task, in order to observe an interesting effect.



Schematic setup

Procedure

The logic gates are connected following the reference schematic. The results observed are unexpected. There's a clear undesired behavior that lasts 40 ns, once the stimulus rises.



Experimental setup



Glitch observation

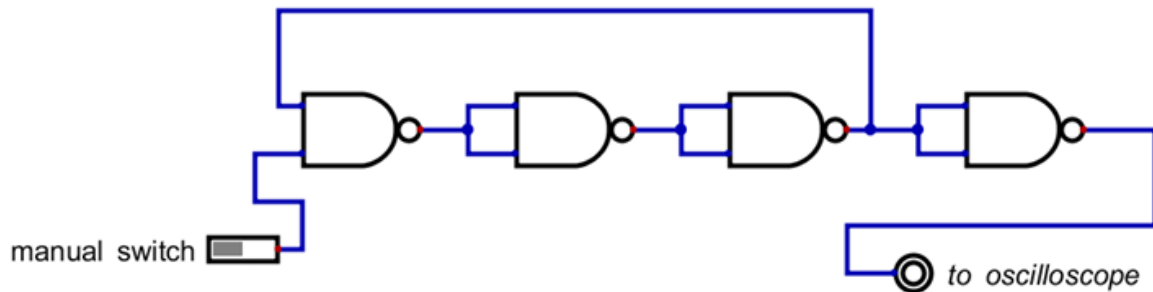
Conclusion

Considering that the last NAND responds to two inputs, one that changes instantly, and one that traverses three extra logic gates, the behavior is clearly understood accounting for these delays between input signals. It takes ~ 30 ns for the second input to reach the final gate, time in which the output is temporarily opposite to the desired state.

Task 4: Combinatorial feedback

Introduction

This exercise introduces the concept of combinatorial feedback and oscillators.



Schematic setup

Procedure

Connect the components as per the schematic, and analyze the output of the system with an oscilloscope for two different cases, when the switch is ON, and when it is OFF.

For a switch setted OFF (case 1), the system response is constantly 1, unaffected by the feedback.

For a switch setted ON (case 2), the system behaves as an oscillator, with values changing between 0/1 and asymmetrical rise and fall times. The period of the resulting signal is 90 ns, resulting in an oscillation frequency of 11,111 MHz (periodic number).



Function generator and oscilloscope, Task 4.



General overview, case 1



Zoomed perspective of the oscillation, case 2

Conclusion

The initial value of the feedback input pin is unpredictable; however, the general behavior of the system is preserved.

On the other hand, the asymmetric rise and fall times of the oscillation are due to the internal PNP and NPN transistors. These are clearly not selected for the purposes of an oscillator, and therefore asymmetrical components result in asymmetrical oscillations. Nonetheless, it is insightful to realize that an oscillator of 11 MHz can be created by the simple means of four logic gates.